

Mayur Kishore Kumar

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SUMMARY

Machine Learning Engineer building production ML and AI systems across semiconductor and healthcare domains. Experienced in predictive modeling, RAG, multi-agent systems, and ML infrastructure, with hands-on experience deploying enterprise AI solutions at scale across production and air-gapped environments.

SKILLS

Languages: Python, SQL, Java, C, C++

Machine Learning: PyTorch, TensorFlow, Scikit-Learn, NumPy, Pandas, XGBoost

Generative AI & LLMs: RAG, AI Agents, MCP, LangChain, LangGraph, Prompt Engineering, Embeddings, FAISS, ChromaDB, Ollama, Groq, n8n

ML Infrastructure & Cloud: Linux, Docker, Kubernetes, MLflow, FastAPI, Flask, AWS, Azure, GCP, PostgreSQL, Git

Computer Vision: YOLO, OpenCV, Image Processing, PaddleOCR, EasyOCR, Tesseract

WORK EXPERIENCE

RAMBUS

Bangalore, India

Machine Learning Engineer Apprentice

Jul 2025 – Present

- **Built an end-to-end EDA resource prediction system** using **IBM LSF** and **Spectre** data, scaling the training dataset from **30K** to **1.5M+** production jobs across **4+** years, evaluating multiple models and achieving an **R-squared of 0.87** for **90th-percentile** resource prediction using **quantile regression**.
- **Designed and deployed an enterprise RAG ecosystem** combining **BM25 keyword retrieval**, **vector search**, **reranking**, and **knowledge-graph retrieval**, with department-specific knowledge bases used across the organization and user feedback indicating **40% average time savings** across AI-assisted workflows.
- **Developed 15 MCP integrations personally (~20 across the team)** for **Jira**, **Confluence**, **Jenkins**, and **ServiceNow** using **Python/FastMCP**, with the infrastructure serving **hundreds of users** and supporting **~30% of relevant ticket activity**.
- **Engineered AI developer tooling and agentic workflows** across **GitHub Copilot**, **Continue.dev**, **terminal-based agents**, and **OpenClaw**, including **70 engineering skills and integrations**, with agent infrastructure supporting **80–100 daily users** and **millions of tokens of usage per day**.
- **Automated engineering workflows** including **NetApp filesystem analytics**, **VS Code extension provisioning**, and **SFTP-based semiconductor data pipelines**.
- **Developed GenAI log-analysis and document-intelligence workflows** using **LLMs**, **OCR**, and **enterprise APIs**, used by **20–30 finance users**.
- **Supported enterprise AI adoption** through **Level 1 and Level 2 training**, helping engineers integrate **AI-assisted development** and **agentic workflows** into day-to-day engineering tasks.

HUMANITARIANS AI

Boston, MA, USA

AI Engineer

Sep 2024 – May 2025

- Created **10+ AI applications from scratch**, building **custom RAG pipelines** and **multi-agent systems** without relying on orchestration frameworks.
- Constructed **end-to-end RAG pipelines** for **clinical triage and patient routing** across **10K+ documents**, engineering the retrieval stack from **document ingestion and semantic search** through **reranking and prompt orchestration**.
- Engineered **multi-agent clinical decision-support workflows** for **patient triage and routing**, combining **planning, memory, and tool orchestration** to coordinate **multi-step reasoning and task execution**.
- Benchmarked **transformer-based language models**, **embedding models**, and **vector databases (FAISS and ChromaDB)** to optimize **semantic retrieval quality**, **inference cost**, and **deployment performance** for **low-resource healthcare environments**.
- Enabled **local (Ollama)** and **cloud-based (Groq)** inference services, allowing **production AI applications** to operate across environments with varying **connectivity and compute constraints**.

AFTERQUERY EXPERTS

San Francisco, CA, USA

Python Engineering Expert (Part Time)

Mar 2025 – May 2025

- **Diagnosed and resolved open issues** across **large-scale public Python GitHub repositories**, implementing **complex solutions** and submitting **production-ready patch files** across multiple codebases.
- **Replicated application environments using Docker** to isolate and debug issues, developing **5+ custom test cases per fix** to validate solutions and ensure **robust error handling**.

INDIAN SPACE RESEARCH ORGANIZATION(ISRO)

Project Trainee

Bangalore, India

Mar 2021 – Jun 2021

- Developed **predictive monitoring models** using **10+ years** of telemetry data to detect **server overheating risks**, enabling proactive **anomaly detection** through monitoring dashboards and alerting workflows that reduced **response time by 30%**.

EDUCATION

NORTHEASTERN UNIVERSITY

Master of Science in Artificial Intelligence - 3.42/4

Boston, MA

Jun 2024

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Bachelor of Engineering in Computer Science - 8.77/10

Bangalore, India

Jul 2022

PROJECTS

MULTI-AGENT AI REASONING PLATFORM

- Developed an extensible **multi-agent reasoning framework** integrating **SQL databases, REST APIs, and document retrieval pipelines** using Groq and Ollama.
- Reduced complex business query response time to **under 10 seconds** through staged reasoning and parallel tool execution.

EXPLAINABLE NBA GAME PREDICTIONS

- Developed an NBA game outcome prediction system achieving **85% accuracy** using **XGBoost** with feature engineering and hyperparameter tuning, while leveraging **BERT, Llama, and Mistral** for natural-language explanation and analysis of model predictions.
- Applied **SHAP** and **LIME** to explain model predictions, visualizing feature importance and decision behavior through Matplotlib and Seaborn to improve model interpretability.